Mathematical Expressions, Arrays and Matrices

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November 11, 2026

1 Introduction

LaTeX provides powerful mathematical typesetting features. It is commonly used for scientific documents, engineering reports, research papers, and mathematical publications.

The amsmath package provides additional environments and commands for writing mathematical expressions.

2 Inline and Display Mathematics

An inline expression appears within a sentence, such as $a^2 + b^2 = c^2$.

A displayed expression appears on a separate line:

$$a^2 + b^2 = c^2$$

3 Fractions and Square Roots

A fraction can be written using the `\frac` command:

$$\frac{a+b}{c+d}$$

A square root can be written using `\sqrt`:

$$\sqrt{x^2 + y^2}$$

4 Subscripts and Superscripts

Subscripts are created using the underscore symbol.

$$x_1, \quad x_2, \quad x_{10}$$

Superscripts are created using the caret symbol.

$$x^2, \quad x^3, \quad x^{10}$$

Both can be combined:

$$x_1^2 + x_2^2$$

5 Greek Letters and Mathematical Symbols

Some commonly used Greek letters are:

$$\alpha,\ \beta,\ \gamma,\ \delta,\ \theta,\ \lambda,\ \pi$$

Other mathematical symbols include:

$$a \leq b, \quad a \geq b, \quad a \neq b$$

$$a \approx b, \quad a \times b, \quad a \pm b$$

6 Summation, Product and Integration

Summation:

$$\sum_{i=1}^{n} i$$

Product:

$$\prod_{i=1}^{n} i$$

Integration:

$$\int_0^1 x^2\, dx$$

Limit:

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

7 Numbered Equations

The equation environment automatically numbers an equation.

$$E = mc^2 \tag{1}$$

$$a^2 + b^2 = c^2 \tag{2}$$

8 Aligned Equations

The align environment aligns multiple equations at the ampersand symbol.

$$x + y = 10 \tag{3}$$
$$x - y = 2 \tag{4}$$

9 Arrays

The array environment arranges mathematical elements in rows and columns.

A	B	C
1	2	3
4	5	6

10 Matrices

A simple matrix:

$$\begin{matrix} 1 & 2 \\ 3 & 4 \end{matrix}$$

A matrix with parentheses:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

A matrix with square brackets:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

A determinant-style matrix:

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$$

11 Conclusion

This program demonstrates mathematical expressions, fractions, roots, Greek letters, summations, equations, arrays, and different matrix environments in LaTeX.